As we have seen, $\sigma_z \otimes I$ acts on Alice's half of the state-vector with σ_z, and does *absolutely nothing* to Bob's half. Because we are working in a four-dimensional vector space, the resulting matrix will be 4 × 4. Omitting multiple ⊗ symbols to avoid visual clutter, we can write the matrix like this:

$$\sigma_z \otimes I =$$

$$\begin{pmatrix} \langle uu|\sigma_z I|uu\rangle & \langle uu|\sigma_z I|ud\rangle & \langle uu|\sigma_z I|du\rangle & \langle uu|\sigma_z I|dd\rangle \\ \langle ud|\sigma_z I|uu\rangle & \langle ud|\sigma_z I|ud\rangle & \langle ud|\sigma_z I|du\rangle & \langle ud|\sigma_z I|dd\rangle \\ \langle du|\sigma_z I|uu\rangle & \langle du|\sigma_z I|ud\rangle & \langle du|\sigma_z I|du\rangle & \langle du|\sigma_z I|dd\rangle \\ \langle dd|\sigma_z I|uu\rangle & \langle dd|\sigma_z I|ud\rangle & \langle dd|\sigma_z I|du\rangle & \langle dd|\sigma_z I|dd\rangle \end{pmatrix}. \tag{7.2}$$

To evaluate these matrix elements, we could allow σ_z and I to operate either to the left or to the right. Let's assume σ_z operates to the left and I operates to the right. Since I does nothing, all we care about is what σ_z does to the bra vector on its left. And within that bra vector, σ_z only acts on the leftmost (that is, Alice's) state-label. Using the rules we've already worked out (see Eqs. 6.6 and 6.7), we can carry out all of these σ_z operations to obtain a matrix of inner products: